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Research Article

Denoising under a data-processing ceiling: fidelity-task divergence in real low-dose CT and in a controlled synthetic matrix

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- Title/Abstract
- Author Information
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- Conflict of Interest and Data Policies
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Title

Denoising under a data-processing ceiling: fidelity-task divergence in real low-dose CT and in a controlled synthetic matrix

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Abstract

Background: Denoising is widely held to improve image quality, evidenced by reference-based fidelity metrics, and thereby to permit dose reduction. The second claim does not follow from the first: processing is a function of the image it is given and cannot add task information to it. Purpose: To quantify when fidelity gains correspond to gains or losses in task detectability, how those effects depend on observer efficiency and on the detectability available in the input, and whether they survive an acquisition nobody controlled. Methods: Two arms. Controlled: signal-known-exactly / background-known-exactly detection of a low-contrast disk on synthetic phantoms over a matrix of dose, noise correlation length and lesion configuration, with three deterministic denoisers (Gaussian, total variation, non-local means) and three observers: prewhitening linear (PW), channelised Hotelling (CHO), and non-prewhitening with an eye filter (NPWE). Each processed arm is referenced to the unprocessed arm of its own condition and to that input's analytic ideal-observer detectability - the ceiling the data-processing inequality and the Neyman-Pearson lemma place on any processing of it. Estimates are cross-fitted, uncertainty is by bootstrap over realisations, and comparisons are Holm-adjusted. The matrix ran over 10 realisations (1,696,000 scored trials). Real data: 12 Siemens liver cases from LDCT-and-Projection-data, vendor reconstructions of the routine and quarter-dose acquisitions, a synthetic lesion of known size and contrast inserted into real parenchyma, and a residual CNN trained on 8 cases and evaluated on the 4 it never saw, with a 15% patch-level validation split inside the training cases, at two configurations spanning 86× in parameters. Results. Controlled: ΔSSIM > 0 and Δd'(PW) < 0. Denoising helped the inefficient observer more than the efficient one: B = Δd'(NPWE) − Δd'(PW) = +1.13. The information-floor hypothesis was partially refuted: below the floor fidelity rose while the task estimate fell, but failure patterns did not concentrate there, and excess lesion-like responses were absent throughout. Real data: across 7 methods the rank correlation between PSNR and d' was ρ = -0.29; the method with the best PSNR ranked 6 of 7 on the task; and 0 arms exceeded the closed-form ceiling in any real-data comparison. A larger configuration of the same architecture - 86× the parameters, trained on more patches for more epochs - improved validation loss and gave the best PSNR in the study, 28.77 dB, while lowering detectability from 5.31 to 4.96. Conclusions. Denoising effects differed by observer, in a direction consistent with differences in prewhitening efficiency; processing improved the estimate for the non-prewhitening observer while leaving the prewhitening observer no better, which would be redistribution of existing information rather than creation of new information. Fidelity gains alone do not establish task preservation, and on real low-dose CT they ranked methods close to inversely to the task. The divergence appeared in both arms and persisted under a substantially larger and longer-trained network of the same architecture.

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Previously Rejected Manuscript

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No

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Criteria for Authorship

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Category

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Special Section

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Keyword

low-dose CT
task-based image quality
model observer

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Areas of Expertise

3 - IM- CT. Phantoms - digital
3 - IM- CT. Quality Control and Image Quality Assessment
3 - IM- CT. Phantoms - physical

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Artificial Intelligence and Machine Learning

Is this article on the topic of artificial intelligence or machine learning?

Yes

The number of training, validation, and test sets are described in the Abstract. The number of input data and output results, along with the type of data (e.g. MRI images, CT images, etc.) are mentioned in the Abstract.

Yes

The stage of development is described in the manuscript Introduction.

No

The data, its source, and data composition are described in detail in the Materials section.

Yes

The details of the machine learning algorithm, including pre-processing and training method, are provided in the Methods section. All major results are accompanied by appropriate tests of statistical significance.

Yes

The innovation, significance, and/or contributions to the field of medical physics are discussed in the Discussion section.

Yes

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Submission

Submitting author attests that this manuscript has not/will not be submitted elsewhere while it is under review by Medical Physics. Please initial to confirm.

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Conflict of Interest

Yes, having read the above statement, there is potential conflict of interest. This has been fully detailed in my cover letter.

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Previously Published Data

No, the results/data/figures in this manuscript have not been published elsewhere

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Data Sharing

The data that support the findings of this study are openly available in [repository name] at [URL], reference number [reference number].

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